

MEMA ERP USER GUIDE

Complete Operational Guide: Transforming an Applicant into an Enrolled Student in Class

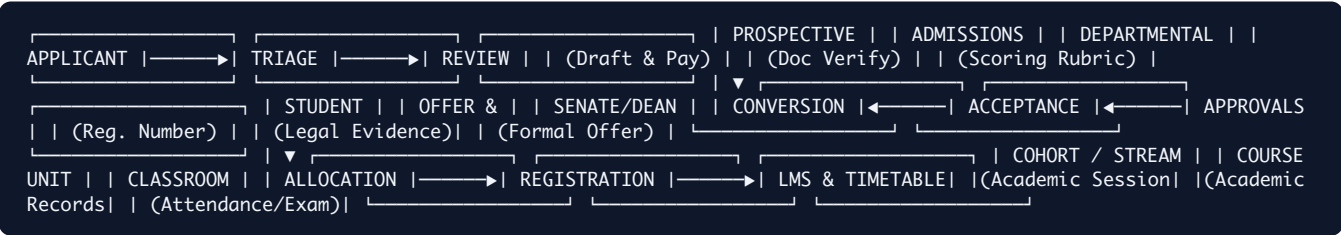
DOCUMENT ID MEMA-UG-ADM-001	SYSTEM VERSION Enterprise v2.4.0	DATABASE & INTEGRITY PostgreSQL • SHA-256 Ledger
KEY STAKEHOLDERS Admissions, Registrar, Finance, Faculty	BASE ACCESS URL http://127.0.0.1:8000	LIFECYCLE STAGES 9 End-to-End Milestones

1. Executive Summary & Core Architecture

The **MEMA ERP Academic Management Suite** executes the applicant-to-student journey through a zero-data-loss, cryptographically verifiable funnel. Unlike legacy systems that duplicate records or drop context upon enrollment, MEMA ERP maintains single-identity continuity: the user who drafts the application retains their audit history, login credentials, and stakeholder permissions when elevated to an official student.

KEY ARCHITECTURAL PILLARS

- 1. Single Identity Continuity:** An applicant account is directly promoted into a student account without creating duplicate login profiles.
- 2. Cryptographic Tamper-Evidence:** Every submission snapshot and offer acceptance is sealed with a SHA-256 evidence hash.
- 3. Atomic Idempotent Conversion:** Database row locks prevent dual student number allocations during concurrent registration surges.



2. Actor Roles & Operational Matrix

Role Key	Portal / Workspace URL	Key Responsibilities
Applicant	/admissions/my-application /applicant/dashboard	Completes application biodata, uploads certificates, pays KES 1,000 application fee, signs offer letter.
Admissions Officer	/admissions/work-queues /admissions/document- verification	Performs intake triage, checks documents for authenticity, requests corrections or marks verified.
Department Reviewer	/admissions/reviews /admissions/shortlists	Evaluates KCSE/qualification clusters against rubric scoring criteria; shortlists or waitlists applicants.
Dean / Registrar	/admissions/approvals /admissions/offers	Authorizes admission batches, signs off letters, sets acceptance deadlines, issues QR verification tokens.
Finance Officer	/fees/payment-receipt /admissions/payments	Reconciles bank/MPESA transactions, confirms tuition fee clearance, applies scholarship waivers.
Academic Registrar	/admissions/student- conversions /registration/student- registrations	Executes the student conversion pipeline, assigns registration numbers, maps cohorts and streams.
Lecturer / Faculty	/curriculum/instructor-mapping /lms/live-lectures	Teaches assigned units, tracks lecture attendance, captures continuous assessment and examination marks.

3. Step-by-Step Lifecycle Guide

STAGE 1 Public Intake & Application Submission

The prospective student accesses `/admissions/my-application`, creates an account, and selects an active intake (e.g. *September 2026 Intake*).

- **Biodata & Academic History:** Candidate completes primary, secondary (KCSE), and tertiary educational records.
- **Document Uploads:** National ID, KCSE result slips, passport photos, and leaving certificates are uploaded to `ApplicationDocument`.
- **Application Fee Payment:** KES 1,000 application fee is validated via MPESA / bank integration (`ApplicationPaymentAttempt`).
- **Cryptographic Sealing:** Upon submission, the system generates a SHA-256 hashed snapshot (`ApplicationVersion`), issues receipt MC-REC-YYYY-XXXXX, and transitions status: `DRAFT` → `SUBMITTED`.

STAGE 2 Admissions Office Triage & Document Verification

Admissions clerks operate from the live work queues at `/admissions/document-verification`.

- **Desk Locking:** Clicking *Claim File* assigns the dossier to the officer, preventing dual-processing.
- **Verification Checks:** Officer cross-checks certificates against national databases and minimum course requirements.
- **Correction Requests:** If a document is defective, the officer triggers `RETURNED_FOR_CORRECTION` or `INFO_REQUESTED` with specific feedback.
- **Advancement:** Satisfactory applications are marked verified: `SUBMITTED` → `VERIFIED`.

STAGE 3 Departmental Scoring, Review & Shortlisting

Academic departments evaluate candidates at `/admissions/reviews` and `/admissions/shortlists`.

- **Scoring Rubric:** Evaluation based on Mean Grade (0–50 pts), Cluster Subjects (0–30 pts), and Affirmative Action/Talents (0–20 pts).
- **Shortlisting:** Applicants meeting cutoff points advance to `SHORTLISTED`.
- **Waitlist Management:** Qualified candidates exceeding current intake quotas are placed on `WAITLISTED` with automatic queue promotion.

STAGE 4 Institutional Approvals & Offer Letter Issuance

The Faculty Dean and Registrar review approval batches at `/admissions/approvals`.

- **Formal Sign-Off:** Batch approval transitions candidates to `ADMITTED`.
- **Automated Offer Generation:** System allocates unique Offer Number (e.g., MC/ADM/202609/A8F19B20) and generates a 48-character QR verification token.
- **Admission Letter PDF:** Generates official university admission package containing reporting dates, campus guidelines, and fee requirements.

STAGE 5 Offer Response, Acceptance & Legal Commitment

The applicant logs into `/applicant/applications/{id}/offer` to review and accept their offer.

- **Forensic Non-Repudiation:** Acceptance captures legal proof: client IP address, browser User-Agent, UTC timestamp, and SHA-256 evidence hash.
- **Status Progression:** Automatic advancement: `ADMITTED` → `ACCEPTED` → `READY_TO_ENROL`.

STAGE 6 Finance Clearance & Fee Reconciliation

The Finance Office validates tuition payments at `/fees/pending-payments` and `/admissions/payments`.

- **Semester Invoicing:** Invoices for Year 1 Semester 1 tuition, caution fees, library levy, and student ID are generated.
- **Payment Verification:** Bank slips / Paybill transactions are matched and confirmed, granting financial clearance.

STAGE 7 Student Conversion & Permanent Identity Materialization

Executed via `/admissions/student-conversions` or action `/admissions/applications/{id}/convert` by the Registrar.

MECHANICS OF STUDENTCONVERSIONSERVICE

`StudentConversionService` performs an atomic database transaction:

1. Obtains a pessimistic lock on the application and course records.
2. Increments `courses.next_student_serial` non-atomically within the lock.
3. Formats permanent Admission Number: BCS/042/2026 (Course / Serial / Year).
4. Inserts permanent row in `students` linked to `academic_sessions`.
5. Upgrades `User.role = 'student'` and adds stakeholder permission.
6. Updates application status to **ENROLLED**.

STAGE 8 Cohort Mapping, Stream Allocation & Unit Enrollment

Academic Affairs links the student to classes and curriculum modules.

- **Cohort Assignment:** Student is linked to Cohort 2026/2027, Stage Year 1 Semester 1, and Stream A (`/cohort/programme-cohort-mapping`).
- **Course Unit Registration:** Student is registered for standard curriculum units (e.g. *Introduction to Programming*, *Discrete Mathematics*, *Computer Architecture*) via `/registration/student-registrations`.

STAGE 9 Classroom Readiness, LMS Sync & Academic Attendance

The student enters the active learning environment.

- **LMS / Moodle Provisioning:** Automated sync provisions course shells, syllabi, and assignment portals (`/lms/course-shells`).
- **Timetable & Attendance:** Classes appear on the student timetable. Lecturers capture live attendance (`AttendanceRecord`).
- **Examinations & Progression:** Student sits continuous assessments, semester exams (`/examination/exam-schedule`), and receives Senate-approved transcripts.

4. Exception Handling & Troubleshooting Directory

Condition / Issue	Root Cause	Resolution Procedure
Application submission disabled	Application fee unpaid or required documents missing.	Verify KES 1,000 transaction on <code>/admissions/payments</code> . Ensure all required upload fields are satisfied.
Expired admission offer	Candidate failed to accept before deadline.	Admissions Officer opens <code>/admissions/offers</code> , edits <code>expires_at</code> timestamp, and re-notifies applicant.
Conversion FAILED in table	Programme offering missing linked <code>course_id</code> .	Navigate to <code>/admissions/student-conversions</code> . Fix course association, then click Retry Conversion .
Course units missing in portal	Programme cohort mapping not published.	Navigate to <code>/cohort/programme-cohort-mapping</code> and assign active curriculum units to the cohort.
LMS courses not accessible	Moodle synchronization job queued.	Open <code>/registration/moodle-sync</code> and trigger manual sync to immediately push registered student shells.

ENTERPRISE GOVERNANCE & AUDIT ASSURANCE

All state transitions, IP addresses, execution channels, and staff decisions are permanently recorded in the live PostgreSQL audit ledger. Access the real-time compliance dashboard at `/reports/audit-trail-user`.